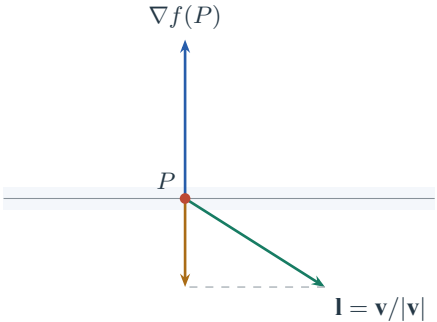
# 方向导数与切平面：负投影对应下降




$$\nabla f(P) = (2, -4, 6)$$
$$\mathbf{l} = (2, 1, -2)/3$$

$$\mathbf{l}_{\max} = \frac{(1,-2,3)}{\sqrt{14}}$$
$$|\nabla f(P)| = 2\sqrt{14}$$

$$D_{\mathbf{l}} f(P) = -4$$
$$\text{切平面：} x - 2y + 3z = 6$$

法截面：切平面为水平线；给定方向含负法向分量